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LAMP-BEAC: Detection of SARS-CoV-2 RNA Using RT-LAMP and Molecular Beacons

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Abstract

SARS-CoV-2 has caused a global pandemic, resulting in the need for rapid assays to allow diagnosis and prevention of transmission. Reverse Transcription-Polymerase Chain Reaction (RT-PCR) provides a gold standard assay for SARS-CoV-2 RNA, but tests are expensive and supply chains are potentially fragile, motivating interest in additional assay methods. Reverse transcription and Loop Mediated Isothermal Amplification (RT-LAMP) provides an alternative that uses alternative and often cheaper reagents without the need for thermocyclers. The presence of SARS-CoV-2 RNA is typically detected using dyes to report bulk amplification of DNA; however a common artifact is nonspecific DNA amplification, complicating detection. Here we describe the design and testing of molecular beacons, which allow sequence-specific detection of SARS-CoV-2 genomes with improved discrimination in simple reaction mixtures. We also show how beacons with different fluorescent labels can allow convenient multiplex detection of several amplicons in “single pot” reactions.

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Introduction

Infection with the beta-coronavirus SARS-CoV-2 causes the disease COVID-19[1]. Transmission typically takes place via airborne respiratory droplets or by contact with contaminated surfaces[2,3]. The virus was identified first in China in December 2019 and later in most countries. On March 11, 2020, The World Health Organization declared COVID-19 a global pandemic. Rapid and efficient detection of SARS-CoV-2 is essential for the control and treatment of this virus.

Numerous methods have been developed to detect SARS-CoV-2 infection. RT-PCR has been widely used to detect SARS-CoV-2 RNA in respiratory specimens[4]. RT-PCR has the advantage of providing accurate and sensitive detection, but supply chain issues have limited testing, motivating the development of additional methods using orthogonal supply chains. RT-LAMP has been widely studied as an alternative[5-8]. In LAMP assays, amplification takes place via a “rolling hairpin” mechanism at a single temperature. Reaction mixtures typically use polymerase enzymes different from those used for PCR, helping avoid supply chain bottle necks. In addition, RT-LAMP can be implemented on neat saliva, or on RNA purified using simple reagents available in bulk[9], again helping bypass supply chain issues and adding robustness to assays.RT-LAMP assays are typically not a sensitive as RT-qPCR assays[10]. However studies suggest that infected individuals are far more infectious during periods of peak viral loads, where RT-LAMP achieves adequate sensitivity. A recent modeling study emphasized that frequency of testing and speed of reporting results are much more important than assay sensitivity, emphasizing

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the value of assays like RT-LAMP that may be implemented efficiently and cheaply[10].

However, a complication is that RT-LAMP reactions often result in non-specific amplification in the absence of target, particularly at longer reaction times, limiting sensitivity. This off-target amplification is especially problematic because LAMP reactions are commonly quantified using colorimetric or fluorescent dyes reporting only bulk DNA synthesis. To counteract these problems, sequence-specific improvements have been proposed based on DNA sequencing[11] or CAS enzymes[12]. These methods are promising, but as presently designed they typically require opening of RT-LAMP tubes and secondary manipulation of reaction products, which has the potential to result in contamination of subsequent reactions with amplification products from previous assays.

Previous research has shown the potential for molecular beacons[13] to allow sequence-specific detection of LAMP products in “single-pot” assays[14,15]. Here, we adapt molecular beacons to detect SARS-CoV-2 sequences, a method we have named LAMP-BEAC (Fig. 1). Molecular beacons are target-specific oligonucleotides labeled with a fluorophore on one end and a quencher on the other. The beacons are designed to incorporate complimentary sequences on their 5’ and 3’ ends such that at low temperatures the ends anneal to form a hairpin, and proximity between the quencher and fluorophore results in decreased fluorescence. When the target of interest is present, the complementary target-specific sequence anneals to its target, separating the

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fluorophore from the quencher and greatly increasing fluorescent signal. The increase in fluorescence resulting from annealing of the beacon probe can be detected without manipulation of the product or opening the reaction tube. Here we describe molecular beacons for detection of SARS-CoV-2 RNA, their use to detect viral RNA in saliva, and combinations of LAMP-BEAC assays for single-pot multiplex detection.

Results

Designing molecular beacons for SARS-CoV-2 RT-LAMP

Several beacons were initially tested for detection of SARS-CoV-2 cDNA in RT-LAMP reactions (Table S1). Optimization required identifying sequence designs that performed properly under the conditions of the RT-LAMP reaction, which is typically run at temperatures around 65°C. Function of the beacon requires that the hairpin remain mostly folded in the hairpin structure at this temperature, while still opening sufficiently often to allow annealing to the target RT-LAMP cDNA product. The annealed beacon-target cDNA duplex must then be sufficiently stable at 65 °C to result in unquenching and an increase in fluorescence. To aid in annealing at these temperatures, we substituted six dNTP positions within the target sequence of the beacon with locked nucleic acids[14]. Several beacons were tested for performance using previously reported RT-LAMP amplicons (Supplementary Table 1).

An example of a successful design is the Penn-LF-beac (Supplementary Table 1). The RT-LAMP amplicon used was first reported by El-Tholoth and

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coworkers at the University of Pennsylvania[8], and is here designated "Penn". The favored beacon was designed to target sequences within the forward DNA loop generated during LAMP; thus the beacon is designated loop forward beacon, Penn-LF-beac.

Testing the Penn/LF-beac LAMP-BEAC system.

The Penn/LF-beac system was then tested on synthetic SARS-CoV-2 RNA (Figure 2). Tests were carried out with commercial LAMP polymerase and reverse transcriptase preparations. In addition, to avoid possible supply chain problems and allow potential production of reagents in resource limited settings, we produced and purified openly available Bst polymerase and HIV reverse transcriptase (described in methods). These enzymes were tested in parallel to the commercial preparations for some tests.

To compare standard LAMP amplification with LAMP-beac, reactions were run containing both the Syto9 fluorescent dye (Figure 2A), which detects bulk DNA by intercalation, and the Penn-LF-beac (Figure 2B). The standard RT-LAMP showed bulk DNA production at shorter times than with the water control, but the water control did amplify shortly after the positive samples. This spurious late amplification is commonly seen with RT-LAMP, though the mechanism is unclear. The primers may interact with each other to form products and launch amplification, or perhaps the reaction is triggered on adventitious environmental DNA. In separate tests, production of products was also shown to depend on addition of LAMP primers (data not shown).

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Reactions detected with the Penn-LF-beac showed more clear-cut discrimination (Figure 2B). The positive samples showed positive signal after about the same amount of time as for the conventional RT-LAMP. However, no signal was detected for the negative water control. Lack of amplification in negative controls has been reproducible over multiple independent reactions (data not shown).

The nature of the products formed could be tested using thermal denaturation (Figure 3). Reactions were first cooled to allow full annealing of complementary DNA strands, then slowly heated while recording fluorescence intensity. The fluorescent signal of the intercalating dye started high but dropped at increased temperatures in all samples (Figure 3A), consistent with denaturation of the duplex and release of the intercalating dye into solution, where it becomes quenched by binding water. In contrast, the beacon's fluorescent signal in the water controls started at low levels of fluorescence (Figure 3B), consistent with annealing of the beacon DNA termini to form the hairpin structure (Figure 1). At temperatures above about 70°C the fluorescence modestly increased, consistent with opening of the hairpin and reptation of the beacon as a random coil. For reactions containing the RT-LAMP product and LF-beac, fluorescence values were high at lower temperatures, consistent with formation of the annealed duplex, then at temperature sufficient for denaturation the fluorescence values fell to match those of the random coil (Figure 3B). Thus the LAMP-BEAC assay generates strong fluorescence signals during LAMP

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amplification in the presence of target RNA but not in negative controls and the thermal melting properties are consistent with the expected products.

LAMP-BEAC assays on saliva.

We then tested performance of the assay on human saliva spiked with a SARS-CoV-2 viral stock. We chose saliva as an analyte because it can be collected simply and inexpensively, and no specialized materials are required that might be subject to supply chain limitations or high costs. A viral stock was added to SARS-CoV-2-negative saliva (collected under IRB protocol #842613), then an inactivation procedure was carried out as described[9]. Briefly, the reducing agent TCEP was added to the sample along with EDTA, and the sample was heated at 95 ^{o}C for five minutes. Aliquots were then diluted from 100- to 1 million-fold into SARS-CoV-2-negative saliva that had been inactivated by the same procedure and viral copy number were estimated by RT-qPCR.

Analysis of the step-wise dilution series was carried out using conventional RT-LAMP with the Penn primers and intercalating dye and with LAMP-BEAC assayed using the Penn-LF-beac molecular beacon (Figure 4). In each reaction, 2 microliters of inactivated saliva were tested without further purification in a total volume of 15 microliters. Amplification with conventional RT-LAMP (Figure 4A) showed DNA synthesis in all samples, with a temporal distinction between the samples containing an estimated 320,000 and 3200 viral copies and the water negative control. The sample with 32 copies overlapped with the negative control. Using LAMP-BEAC, the samples containing 320,000 and 3200 genome copies showed amplification, but there was no amplification in

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the sample with 32 copies or the water controls. Melt curve analysis of the LAMP-BEAC products was consistent with the expected molecular structures as described above. Thus LAMP-BEAC was able to detect 3200 copies of viral stock in inactivated saliva without need for RNA purification.

Multiplexing multiple LAMP-BEAC reactions in single reaction tubes

We next investigated whether multiple LAMP-BEAC assays could be multiplexed in single tubes (Figure 5). To test this, we combined two beacons. The first was the beacon targeting the Penn amplicon, Penn-LF-beac, labeled with the fluorescent dye hexachlorofluorescein (Hex), allowing read out of fluorescence at 587 nm. The second beacon, As1e-LB-beac, recognizes the SARS-CoV-2 As1e amplicon reported in[9]. The Ase1-LB-beac beacon is conjugated to the dye cyanine-5 (Cy5), allowing read out of the fluorescent signal at 682 nm. All sequences are presented in Supplementary Table 1. Reactions were carried out with known quantities of the synthetic Twist RNA template. Analysis of the production of bulk DNA was also queried in the same reaction tubes by addition of the Syto9 dye and assayed at 520 nm wavelength.

Bulk DNA amplification was detected in all positive samples prior to amplification in negative controls (Figure 5A). Analysis with either the Penn LF/LB beacons at 587 nm (Figure 5 B and C), or the As1e LB-Cy5 beacon at 682 nm (Figure 5 D and E), showed clear amplification of the higher concentration samples. Melt curve analysis showed changes in signal with temperature as expected for formation of proper products (Figure 5 C and E). Assays

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consistently detected 125 copies of the analyte, and sporadically detected lower amounts. Negative controls were consistently negative, suggesting that the amplification seen with traditional RT-LAMP was nonspecific.

Reactions in Figure 5 A-E were carried out with commercial preparations of the RT-LAMP polymerases--Figure 5 F-J show a similar collection of reactions performed with alternative enzymes manufactured locally as described in the Materials and Methods. In these assays, there was considerable DNA synthesis in the water controls (black lines, Figure 5F). However, the water controls were negative when assayed using the beacons (Figure 5 G-J). Assays consistently detected 250 copies of the test RNA, and often lower. Thus implementation of the multiplexed LAMP-BEAC assay is not reliant on commercial sources of reverse transcriptase and DNA polymerase. These enzymes show frequent non-specific amplification of negative controls in the assay of bulk DNA (Figure 5F) perhaps due to amplification at low temperatures being blocked by a “warm start” aptamer present in the commercial enzyme preparations. However, these nonspecific amplifications did not generate fluorescent signals for the Penn-LF-beac or As1e-LB-beac.

Discussion

Standard RT-LAMP is an attractive method for assay of SARS-CoV-2 RNA in patient samples due to the simplicity of the method and the use of a supply chain orthogonal to the clinical assay supply chain. However, LAMP

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typically detects only the presence of amplified DNA , and thus assays are often complicated by nonspecific amplification. Improved specificity can likely be achieved by sequence-specific detection, and multiple methods have been proposed[11,12]. However, some of these approaches are complicated by the need to open reaction tubes and manipulate products, creating severe danger of contamination between runs. Here we introduce a convenient method for sequence-specific detection of SARS-CoV-2 RNA using molecular beacons that does not require manipulation of reaction products, which we have named LAMP-BEAC.

The LAMP-BEAC method is not as sensitive as RT-QPCR, but it can be implemented inexpensively, potentially allowing frequent assays of vulnerable populations. The reaction set up and incubation can be done in a couple of hours, allowing rapid turn around. Thus the LAMP-BEAC assay potentially meets the needs articulated by modeling studies for effective surveys of asymptomatic populations[10]. To this end, we demonstrate here that LAMP-BEAC works efficiently on inactivated saliva, providing an easily-collected analyte.

LAMP-BEAC is also well suited to multiplexing. Here we show use of two molecular beacons to detect specific sequences in two different amplicons, providing internal validation for positive tests. Going forward, it will be useful to add a human RNA amplicon to verify sample integrity. Additional respiratory viruses could also be assayed, potentially all in one reaction tube. Optimization of such methods are currently under way.

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Methods

Design of Molecular Beacons

Beacons were designed based on previously published LAMP primer sets. To design beacons targeting the loop region of the LAMP amplification product, we mapped the FIP and BIP primers to the SARS-CoV-2 genome to find the entire forward and backward loop regions of the amplicon (potentially including regions outside the original LF and LB primers). We then selected GC-rich subsequences within these loops and selected bases for LNA modification based on the predicted change in melting temperature using a stepwise greedy heuristic of consecutively adding the LNA with the highest predicted Tm. Additional nucleotides were then added to the 5' and 3' ends to form a hairpin with predicted melting temperature between 60-65 ºC. Where possible terminal bases of the target sequence were used as part of the hairpin. To allow easy and relatively affordable synthesis, beacons were kept shorter than 25 nt with 6 locked nucleic acids.

Design and purification of polymerases

We chose the D422A mutant of the polA large fragment from *Geobacillus stearothermophilus* for LAMP due to its high strand displacement activity[16]. Using strain DSM 13240, the polymerase coding sequence was amplified from genomic DNA, the D422A substitution was incorporated, and the construct was ligated into CDFDuet (Novagen) in-frame with an N-terminal hexahistidine tag. The polymerase was expressed in strain BL21(DE3) at 37 °C for 3 h and purified

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using Talon (Clontech), and heparin sepharose (GE) chromatography. Purified polA LF was concentrated, glycerol added to 10%, and aliquots were flash frozen and stored at -80 °C.

To demonstrate that RT-LAMP can be performed using a reverse transcriptase generated in-house, we constructed a synthetic gene for the HIV1 RT p66 subunit containing substitutions expected to confer thermal stability. The p66 sequence was inserted into pET29b and the p51 subunit coding sequence was amplified by PCR and inserted in frame to an N-terminal hexahistidine tag in CDFDuet. The p51 and p66 subunits were co-expressed in BL21(DE3) at 20° for five hours and purified using Talon and heparin sepharose chromatography. Purified RT was concentrated, glycerol added to 10%, and aliquots were flash frozen and stored at -80°C.

RT-LAMP reaction mixtures

RT-LAMP reactions were prepared by mixing 7.5 ul commercial 2x LAMP master mix (NEB E1700L) or our own LAMP mix (40 mM TrisHCl, pH 8.5, 20 mM $(NH4)_2SO4$, 100 mM KCl, 16 mM MgSO4, 0.2% Tween-20, 2.8 mM each dNTP, 16 µg/ml polA LF, and 5 µg/ml RT) with 1.5 ul of 10x primer/beacon master mix (final concentration: 1.6uM FIP/BIP, 0.2 uM F3/B3, 0.4 uM LF/LB, 0.05 uM beacon) and 6ul of sample and/or water. For multiplexed LAMP reactions, the final total concentration of primers/beacons was maintained e.g. the individual

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primer/beacon concentrations were halved when two primer sets were added to the same reaction.

Assay using LAMP-BEAC

LAMP-BEAC reactions were performed at 63-65 ºC with fluorescent quantification every 30 seconds on a QuantStudio 5. Reactions typically completed within 30 minutes but for research purposes data was collected for additional time spans. After reaction completion, the reaction was heated to 95 ºC for 5 minutes to inactivate any remaining enzyme, cooled to 25 ºC (at a rate of 0.1 ºC/sec) and then slowly heated to 95º C with fluorescence measured every degree.

Acknowledgements

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Figure Legends

Figure 1. RT-LAMP assayed using molecular beacons: LAMP-BEAC. The molecular beacon used is shown at the top in the annealed hairpin form, which is quenched. Binding of the beacon to the target complementary sequence separates the fluorescent group and the quencher, allowing detection of

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fluorescence. The red loops on the beacon indicate locked nucleic acids used to increase binding affinity.

Figure 2. Reaction progression curves comparing RT-LAMP assayed using an intercalating dye and LAMP-BEAC. A) Conventional RT-LAMP assay detecting SARS-CoV-2 RNA. Time after reaction initiation is shown on the x-axis, relative fluorescence intensity is shown on the y-axis. The RNA used was the Twist commercial positive control; copy numbers are shown in the key at the bottom. B) Detection of SARS-CoV-2 RNA using LAMP-BEAC. Markings are as in A).

Figure 3. Thermal melting curves to characterize amplification products. The results shown are for reactions in Figure 2; the key to samples tested is at the bottom. Reaction products were cooled to room temperature, then slowly heated for the melt curve analysis. A) Characterization of products generated using conventional RT-LAMP and the Twist RNA template. The x-axis shows the temperature, the y-axis shows fluorescence intensity. B) Characterization of products generated using LAMP-BEAC. Markings as in A).

Figure 4. Assays of saliva samples doped with SARS-CoV-2 particles. Estimated viral copies in the samples analyzed are enumerated in the key at the bottom. A) Detection of reaction products using an intercalating dye. B) Detection of reaction products using LAMP-BEAC (Penn amplicon with LF-beac beacon). C) Melt curve analysis of the reaction products in B).

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Figure 5. Multiplex LAMB-BEAC. Reactions A)-E) are identical to reactions F)-J), except that reactions A)-E) were carried out with commercial RT and DNA polymerase enzymes, while F)-J) were carried out with locally manufactured enzymes. All assays contained the Penn amplicon detected with beacons Penn-LF conjugated to HEX, as well as a published amplicon As1e detected with a beacon As1e-LB-beac conjugated to Cy5. Reaction mixtures also contained the Syto9 intercalating dye. A) and F): detection of reaction products using Syto9 fluorescence (read at 520 nm). B) and G): detection of Penn amplification reaction products using the Penn-LF-beac system (read at 587 nm). C) and H): melt curve analysis read out at 587 nm. D) and I): detection of As1e reaction products using the As1e-B-beac system (read at 682 nm). E) and J): melt curve analysis read out at 682 nm.

Supplementary Material

Table S1. Oligonucleotides used in the LAMP-BEAC assay.

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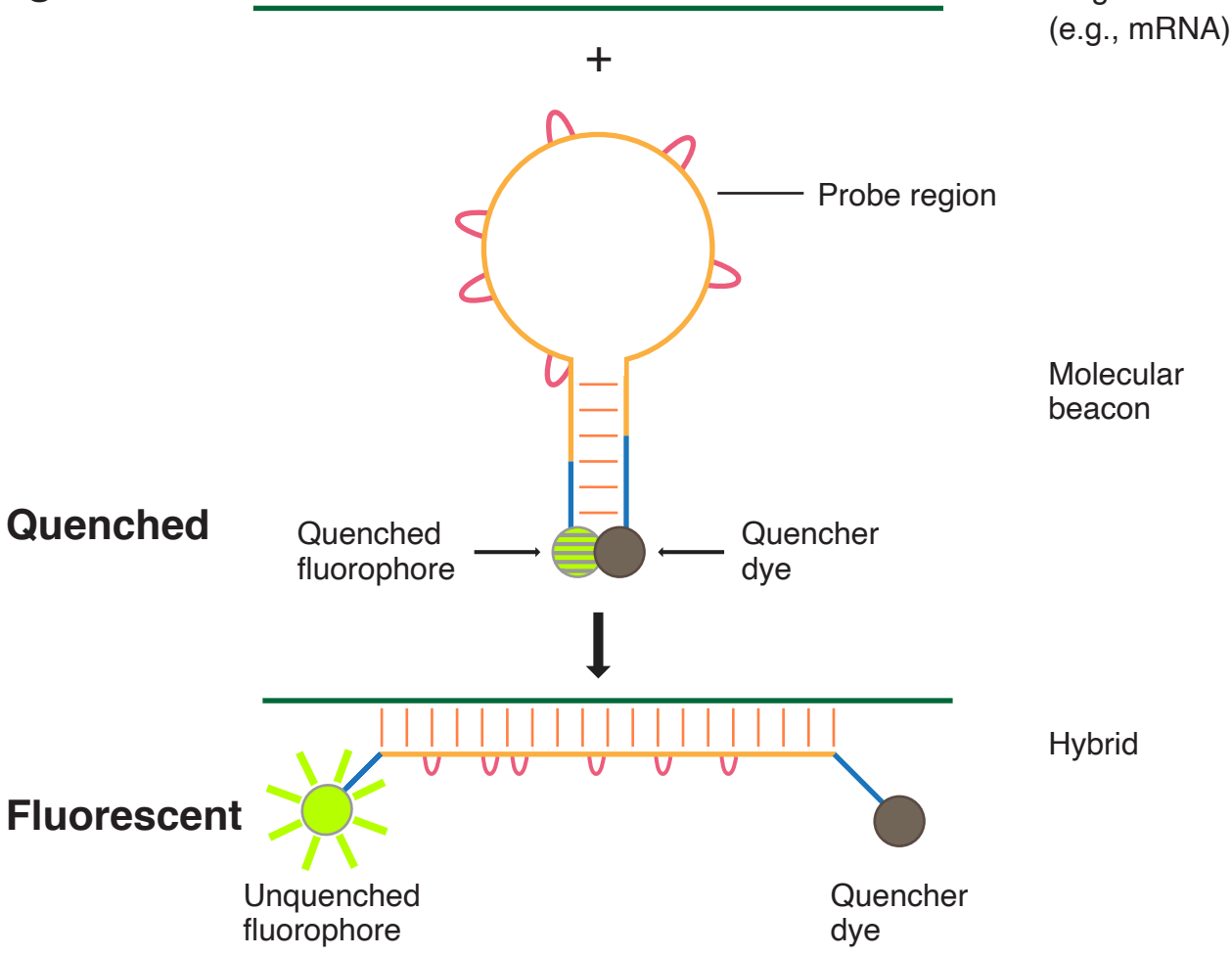
Figure 1


Target
(e.g., mRNA)
+
Probe region
Molecular
beacon
Quenched
Quenched
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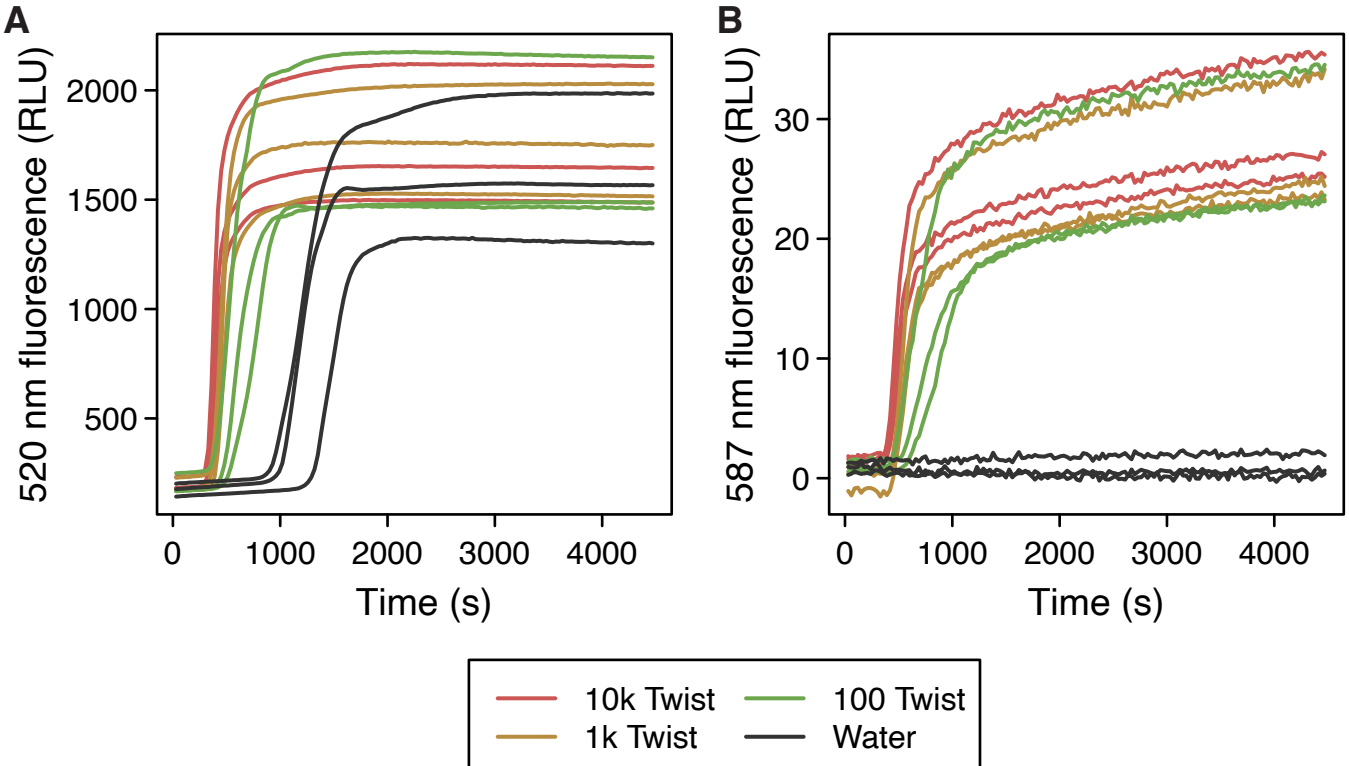
Quencher
dye
Hybrid
Fluorescent
Unquenched
fluorophore
Quencher
dye

Figure 2


A
520 nm fluorescence (RLU)
2000
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Time (s)
B
587 nm fluorescence (RLU)
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10k Twist
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1k Twist
Water

Figure 3

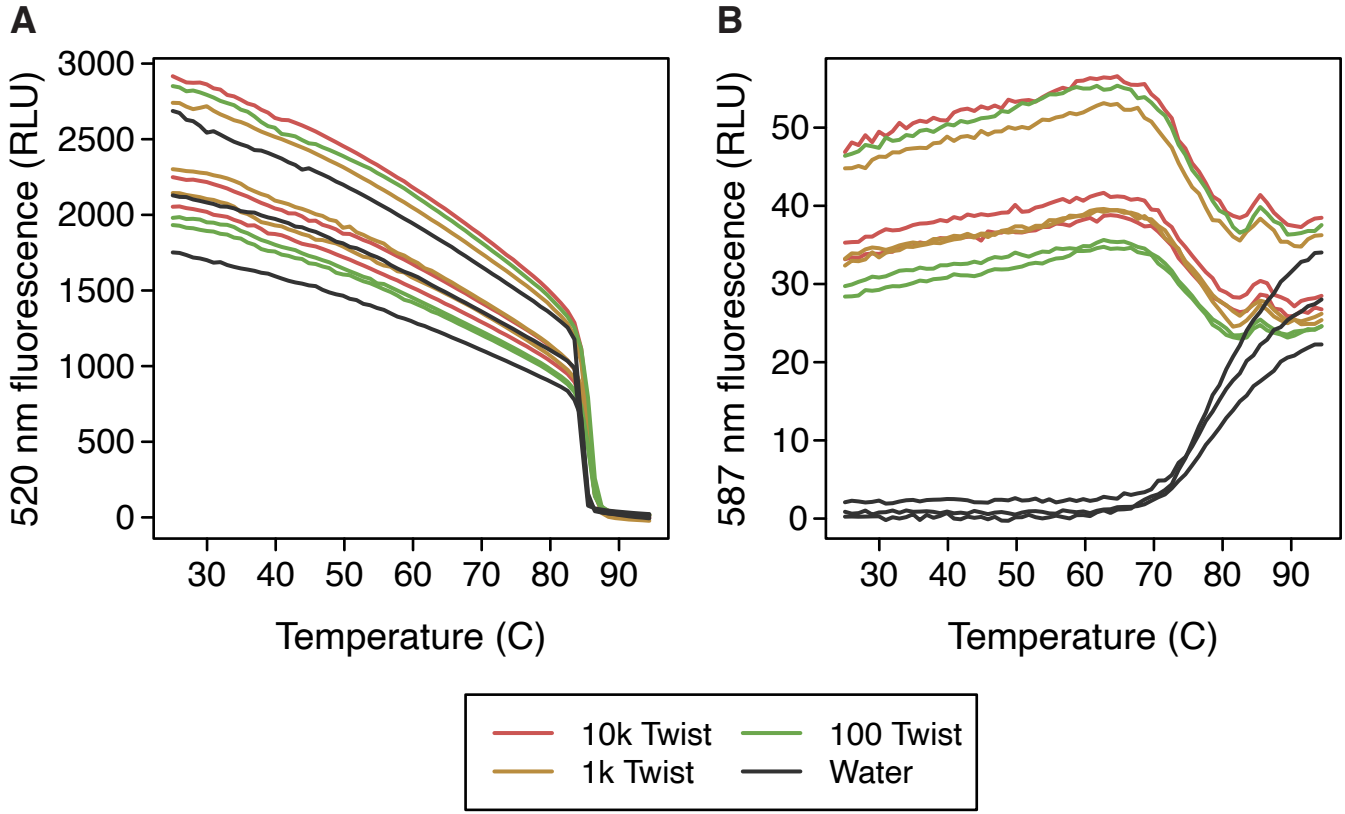
A
B
520 nm fluorescence (RLU)
3000
2500
2000
1500
1000
500
0
30
40
50
60
70
80
90
Temperature (C)
587 nm fluorescence (RLU)
50
40
30
20
10
0
30
40
50
60
70
80
90
Temperature (C)
10k Twist
100 Twist
1k Twist
Water

Figure 4

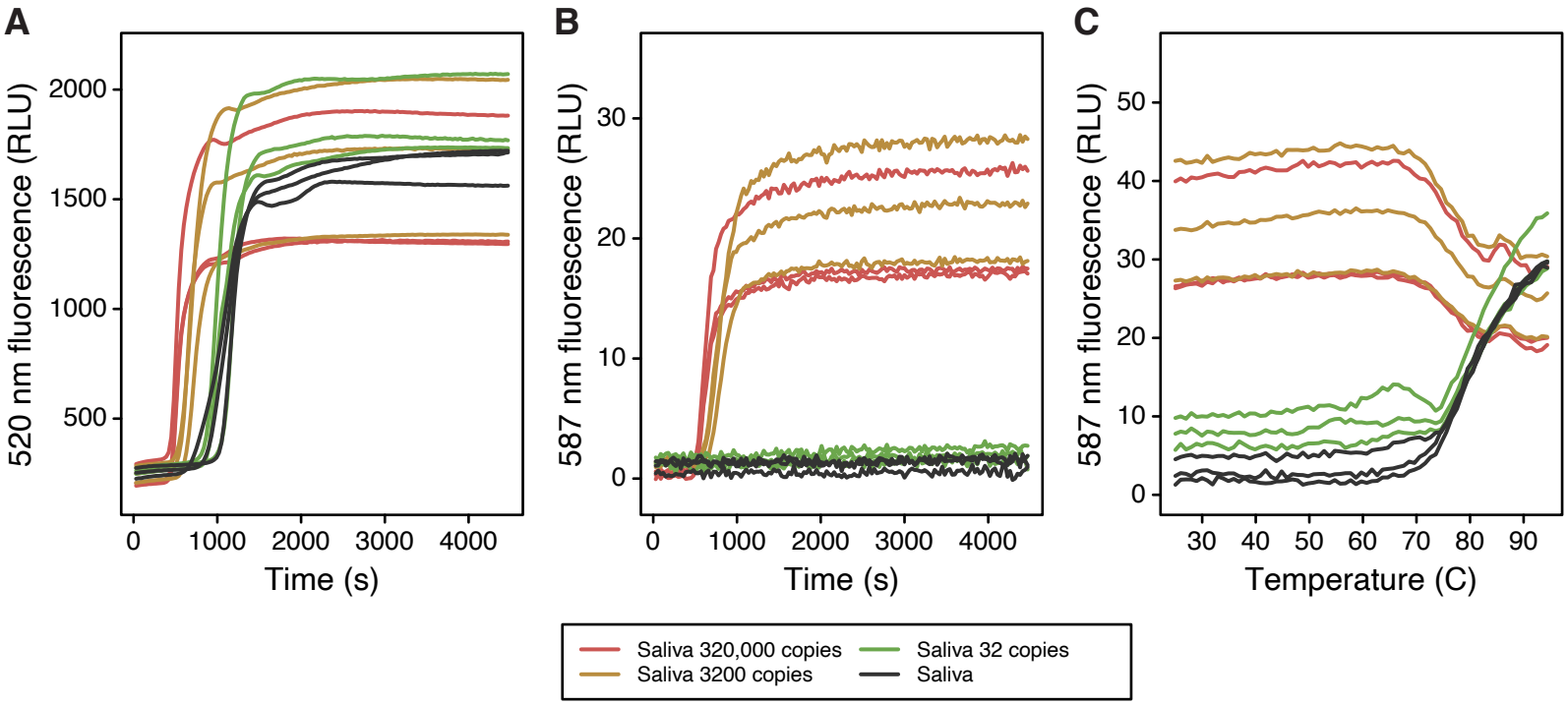
A
520 nm fluorescence (RLU)
2000
1500
1000
500
0
1000
2000
3000
4000
Time (s)
B
587 nm fluorescence (RLU)
30
20
10
0
0
1000
2000
3000
4000
Time (s)
C
587 nm fluorescence (RLU)
50
40
30
20
10
0
30
40
50
60
70
80
90
Temperature (C)
Saliva 320,000 copies
Saliva 3200 copies
Saliva 32 copies
Saliva

Figure 5

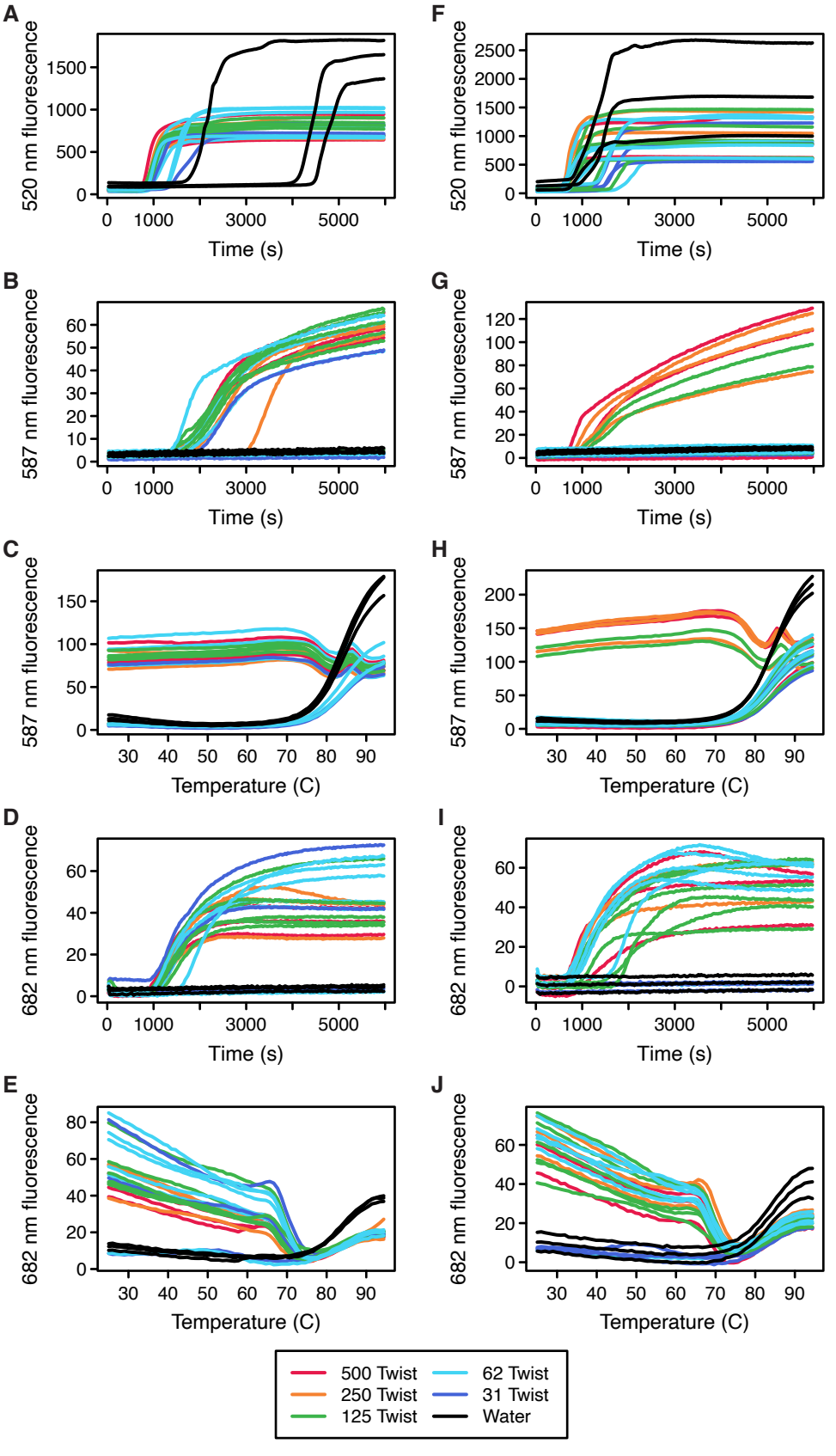
A
520 nm fluorescence
Time (s)
B
587 nm fluorescence
Time (s)
C
587 nm fluorescence
Temperature (C)
D
682 nm fluorescence
Time (s)
E
682 nm fluorescence
Temperature (C)
F
520 nm fluorescence
Time (s)
G
587 nm fluorescence
Time (s)
H
587 nm fluorescence
Temperature (C)
I
682 nm fluorescence
Time (s)
J
682 nm fluorescence
Temperature (C)
500 Twist
250 Twist
125 Twist
62 Twist
31 Twist
Water

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Table S1: Oligonucleotides used in this study

Primer set	Oligo name	Sequence	Source
As1e	As1e_FIP	TCAGCACACAAAGCCAAAAATTTATTTTTCTGTGCAAAGGAAATTAAGGAG	Rabe & Cepko 2020
As1e	As1e_BIP	TATTGGTGGAGCTAAACTTAAAGCCTTTTCTGTACAATCCCTTTGAGTG	Rabe & Cepko 2020
As1e	As1_F3	CGGTGGACAAATTGTCAC	Rabe & Cepko 2020
As1e	As1_B3	CTTCTCTGGATTTAACACACTT	Rabe & Cepko 2020
As1e	As1_LF	TTACAAGCTTAAAGAATGTCTGAACACT	Rabe & Cepko 2020
As1e	As1_LB	TTGAATTTAGGTGAAACATTTGTCACG	Rabe & Cepko 2020
As1e	As1_LBMB_LNA_S1	CTGCGGTGAAA+C+ATT+T+GT+CACG+CAG	This paper
Penn	Penn_FIP	TCAGTACTAGTGCCTGTGCCCACAATCGTTTTTAAACGGGT	El-Tholoth et al. 2020
Penn	Penn_BIP	TCGTATACAGGGCTTTTGACATCTATCTTGGAAGCGACAACAA	El-Tholoth et al. 2020
Penn	Penn_F3	TGCTTCAGTCAGCTGATG	El-Tholoth et al. 2020
Penn	Penn_B3	TTAAATTGTCATCTTCGTCCTT	El-Tholoth et al. 2020
Penn	Penn_LF	CTGCACTTACACCGCAA	El-Tholoth et al. 2020
Penn	Penn_LB	GTAGCTGGTTTTGCTAAATTCC	El-Tholoth et al. 2020
Penn	Penn_LFMB_LNA_S1	CCGGTG+TA+A+GA+CG+GG+CTGCACCGG	This paper
Penn	Penn_LBMB_LNA_S1	CCGCAAAG+T+AG+C+T+G+GTTTTGCGG	This paper